

 UPNM National Defence University of Malaysia Kewajipan • Maruah • Integriti	JABATAN SAINS KOMPUTER FAKULTI SAINS DAN TEKNOLOGI PERTAHANAN UNIVERSITI PERTAHANAN NASIONAL MALAYSIA
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TSK 3306 (PROJEK)

CADANGAN TAJUK PROJEK SARJANA MUDA

Semester I, Sesi Akademik 2025/2026

Arahan: Sila lengkapkan dan hantarkan borang ini kepada Penyelaras PSM. Borang cadangan tajuk projek ini perlu disahkan oleh penyelia.

BAHAGIAN A: MAKLUMAT PELAJAR

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BAHAGIAN B: MAKLUMAT PROJEK

Nama Penyelia:	Prof. Madya Dr. Muslihah binti Wook
Tajuk Projek:	Sentiri: Sentiment Analysis and Topic Modeling System for Autism Therapy Centers Using XLM-RoBERTa

Pernyataan Masalah:

In recent years, Autism Spectrum Disorder (ASD) has increased significantly, creating a very high demand for effective therapy and clinical management solutions. Autism therapy centers face various challenges in managing client data, tracking clinical progress, and analyzing therapy outcomes. Traditional methods often rely on manual documentation, which is really going to make a huge time consumption, tend to produce errors, and the critical one is lacks of analytical capabilities to conclude really meaningful insights from each therapy sessions.

Therapist and clinicians struggle to systematically track client's emotional and behavioral progress over time. Conversations between therapists and clients, which contain valuable clinical insights, are also often stored in unstructured text data without proper analysis. This results in missed opportunities to identify patterns. Moreover, the lack of an integrated platform that combines clinical documentations and sentiment analysis leads to fragmented workflows and reduced operational efficiency.

Therapy centers often conduct session via WhatsApp, where conversation happen in both one-on-one chats between therapist and clients, as well as group chats within the therapy center. These WhatsApp conversation contain rich clinical data that, if we properly analyzed it could be provide such a valuable insight for client. However we see this problem is, extracting and analyzing all of these conversations manually is such a time consuming and impractical. The exported .txt files from WhatsApp contain unstructured data, making it difficult to identify which messages belong to which client and track individual progress over time.

Furthermore, clients are identified by their phone numbers in WhatsApp chats. The system must be able to detect and map these phone numbers to registered clients in the database to ensure that sentiment analysis and progress tracking are accurately associated with the correct individual. Without proper client identification, the analysis becomes meaningless and cannot be used for clinical decision-making.

Despite the availability of various sentiment analysis tools, most are designed for English-language content and fail to handle the linguistic diversity found in Malaysian therapy settings. Therapists and clients in Malaysia often communicate in Bahasa Malaysia, English, and Manglish (Malaysian colloquial English), creating a unique multilingual challenge that existing solutions do not adequately address. This limitation prevents accurate sentiment detection, crucially it will also reducing the quality of clinical insights available to therapists and doctors.

Furthermore, while sentiment analysis reveals how a client feels, it does not explain what they are talking about. Therapists need to know not only the emotional state of their clients but also the current themes and topics that involved during every therapy sessions. That's why I try to implement topic modelling capabilities, clinicians may miss critical patterns in client conversations, such as anxiety, food, family issues or any progress in social skills development.

Existing systems in the market either focus solely on clinical documentation (such as electronic health records) or on basic sentiment analysis, but rarely intergrate both functionalities with topic modelling into a unified platform. Also, many solutions lack of role-based access control to accommodate different user types (Admin, Therapist, Doctor, Client). This fragmentation leads to inefficient workflows, data silos and very limited analytical capabilities for therapy centers.

Hence, this shows that there is a clear need for a comprehensive, integrated platform that empowers autism therapy centers to:

1. Automatically extract and process WhatsApp chat transcripts from .txt export files
2. Detect and map the phone numbers to registered clients in the database
3. Analyze conversations using multilingual sentiment analysis and topic modeling
4. Track individual client progress over time based on their conversation patterns

The Sentiri system is designed to address these gaps by providing an intelligent, all-in-one solution that combines clinical management, WhatsApp chat extraction, multilingual sentiment analysis, topic modelling using XLM-RoBERTa, and administrative oversight in a secure and user-friendly web-based platform.

Objektif Projek:

- a. To design a multilingual sentiment analysis and topic modelling model for autism therapy centre feedback using XLM-RoBERTa.
- b. To develop Sentiri, a system for analysing and visualising sentiments and topics from feedback received by autism therapy centres.
- c. To test and evaluate the functionality of Sentiri and the performance of its sentiment analysis and topic modelling components.

Skop Projek:

1. The development of this system involves four categories of users:
 - a. **Admin:** Responsible for managing the entire system, including user registration, system configuration, and overseeing all clinical and administrative data. The admin has full access to view and manage clients, therapists and doctors. Also responsible for uploading WhatsApp chat transcripts from main group.
 - b. **Therapist:** Responsible for uploading WhatsApp chat transcripts (.txt files), analyzing sessions outcomes, and tracking client progress. Therapists can view only their assigned clients and monitor their sentiment trends and topic patterns over time.
 - c. **Doctor:** Responsible for diagnosing clients with autism and assigning DSM-5 support levels (Level 1,2,3). Doctors have diagnostic authority to add and update clinical data.
 - d. **Client:** Can view their own diagnosis, support level, sentiment analysis results, and topic insights. Clients can track their own progress in a read-only dashboard.
2. WhatsApp Chat Processing will include the following capabilities:
 - a. **File Upload:** Admin and therapists can upload .txt files exported from WhatsApp (both one-on-one group chats).
 - b. **Message Parsing:** The system will parse the .txt file to extract individual messages timestamps and sender identifiers such as names or phone numbers.
 - c. **Phone Number Detection:** The system will automatically detect phone numbers from message headers and map them to registered clients in database.
 - d. **Client Matching:** If a phone number matches a registered client, the message will be linked to that client's profile for analysis and progress tracking.
 - e. **Unmatched Messages:** Messages that cannot be linked to any registered client will be stored separately for manual review by admins or therapists.
 - f. **Batch Processing:** Multiple chat files can be uploaded and processed in batched with unique batch IDs for tracking.

3. Role-based access control (RBAC) will be implemented to ensure that each user type has appropriate permissions and data access levels, maintaining security, data security, data integrity, and compliance with privacy regulations.
4. This application will be developed using the following technologies:
 - a. **Django Framework (Python):** Backend development and database management using Django's ORM and authentication system.
 - b. **Tailwind CSS:** Modern, responsive user interface design with utility-first CSS framework for consistent styling.
 - c. **SQLite/PostgreSQL:** Database management for storing client data, clinical records, chat transcripts and analysis results.
 - d. **XLM-RoBERTa:** Fine tuned multilingual language model for sentiment analysis and topic modeling, capable of processing Bahasa Malaysia, English, and Manglish text.
 - e. **Transformers Library (Hugging Face):** Implementation of XLM-RoBERTa model loading, tokenization, and inference for NLP tasks.
 - f. **Scikit-learn:** Topic modeling and clustering algorithms for identifying themes from XLM-RoBERTa embeddings.
 - g. **Python's Built-in Modules:** For parsing WhatsApp **.txt** export files (datetime, regex for phone number detection).
 - h. **Django Admin Panel:** Built in administrative interface for managing users and models.
5. The system will include the following core modules:
 - a. **User Management Module:** Registration, authentication, profile management, and role-based access control.
 - b. **WhatsApp Chat Import Module:** Upload, parse, and process **.txt** files with phone number detection and client matching.
 - c. **Sentiment Analysis Module:** Automated sentiment detection (positive, negative, neutral) from therapy session transcripts using XLM-RoBERTa.
 - d. **Topic Modeling Module:** Identification of recurring themes and topics from conversation data using XLM-RoBERTa embeddings and clustering techniques.
 - e. **Reporting Module:** Interactive dashboard showing client progress, sentiment trends, and topic insights for therapists and clients.
6. The system will be designed with a focus on:
 - a. **Data Security:** using csrf token for every session
 - b. **Usability:** intuitive user interfaces designed for different user roles and skill levels.
 - c. **Multilingual Support:** Ability to process and analyze therapy conversations in Bahasa Malaysia, English, and Manglish using XLM-RoBERTa.
 - d. **Scalability:** Modular architecture allowing for future feature additions and system expansion.

Konsep Projek:

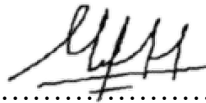
Sentiri is a sentiment analysis and topic modelling system developed to analyse feedback from parents and caregivers regarding services provided by autism therapy centres. The system uses XLM-RoBERTa to process multilingual textual feedback, particularly in English and Bahasa Melayu, and classify the feedback into positive, neutral, and negative sentiments. In addition, topic modelling is used to identify the main issues and themes discussed in the feedback, such as therapy effectiveness, therapist performance, child progress, communication, facilities, appointment management, and service quality. The analysis results are presented through a dashboard that summarises sentiment distributions, key topics, recurring concerns, and feedback trends. Sentiri aims to help autism therapy centres better understand parents' experiences, identify areas that require improvement, and support more informed decision-making in enhancing the quality of their services.

BAHAGIAN C: PENGESAHAN PENYELIA

Saya PROF. MADYA DR. MUSLIHAH BINTI WOOK mengesahkan bahawa saya telah menyemak cadangan projek pelajar ini dan bersetuju untuk mengemukakan cadangan projek ini untuk dinilai.

Tarikh : 10/8/2026

Tandatangan & Nama Penyelia :



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BAHAGIAN D: UNTUK KEGUNAAN JKPSM

Tajuk dilulus penuh ()

Tajuk dilulus bersyarat ()

Tajuk ditolak ()

Nota (*Sila nyatakan syarat atau sebab tajuk ditolak*)

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(Jika tajuk ditolak : Sila berbincang dengan penyelia dan serahkan tajuk PSM yang baru seminggu selepas tarikh penolakan , kepada Penyelaras PSM)

Tandatangan & Nama Penyelaras PSM:.....

Tarikh Mesyuarat JKPSM :.....